Chaitanya Tripathi, Ashoka University, Sonipat, Haryana 131029, India
ORCID 0009-0002-4094-0323 · chaitanya.tripathi_ug2025@ashoka.edu.in
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Professor Peter E. Andersen, Editor-in-Chief, *Applied Optics*

Dear Professor Andersen,

I am submitting "Media-in-the-loop holography: designing photopolymer exposures through a differentiable recording model" for consideration as a Research Article.

A laboratory recording a volume hologram in a photopolymer must choose the exposure before the medium distorts the target, and can measure the error only once it is permanent; camera-in-the-loop correction, which solves the equivalent problem on a spatial light modulator, is unavailable. The manuscript asks what is gained by designing the exposure through a model of the recording instead, and answers three practical questions: when it is worth doing, when it stops working, and what has to be measured first. The gain over a media-blind design is positive in 29 of 30 grid cells, up to 2.52 dB, and removes every exposure falling more than 2 dB short of an oracle reference, which is the benefit that matters when each attempt consumes a film. Section 6 is a step-by-step calibration protocol with the minimum measurement set and the conditions for not using the method; Section 4.6 states the geometric boundary and identifies its cause. The work fits *Applied Optics* in both of the ways the journal asks for: it gives design guidelines, and it explains the limits of implementing theory.

Previous submission, disclosed in full. An earlier version was submitted to *JOSA A* as manuscript ID 620231 and declined on 15 October 2026 by the Editor-in-Chief at the editorial stage, on the grounds that the research area lies outside that journal's direct scope and that *JOSA A* restricts papers relying on computer-simulated results. It was not sent for peer review, no reviews were returned, and no scientific objection to the work was raised. I can supply the decision letter.

What has changed since. The manuscript now quantifies which recording parameters the gain depends on, by two independent routes that agree, and finds the medium's index budget dominating the other three by more than a factor of three, with a several-fold underestimate reversing the benefit. It explains the failure at nonzero fringe slant rather than reporting it: a rigorous coupled-wave sweep over intermediate angles shows the first diffracted order collapsing as the grating detunes from Bragg, leaving the reconstruction to the undiffracted beam, where no exposure design can help. That sweep also shows the closed-form readout reference losing validity at the smallest nonzero angle tested, so the apparent improvement at small slant is reported as model-dependent rather than as an effect. The main text is self-contained; Supplement 1 keeps only full derivations and extra sweeps.

The limitations are stated plainly. The study is in simulation; no hologram has been recorded. The model is compared with published dose-response data for one commercial photopolymer at one spatial frequency, and the manuscript says why that is not enough to establish transfer across spatial frequency or across media, making that measurement the first item of the protocol it recommends to others. Every number is generated by script from released per-job result files, and the code and raw results are public.

The manuscript is original, is not under consideration elsewhere, and runs to 9 pages in the two-column journal layout, with five figures, three tables and a supplemental document. Part 1, a companion study on phase-only computer-generated holography with idealized modulators, is a cited Optica Open preprint and does not overlap with these results. I declare no conflicts of interest and received no external funding; use of a generative AI assistant is disclosed in the Acknowledgment, as Optica policy requires. Suggested reviewers are entered in Prism.

Thank you for considering the manuscript.

Sincerely,
Chaitanya Tripathi